# Chat Transcript

October 4, 2026

**USER**

Run the cell below and copy the printed table.

```
[17]: print(flagged[["report_id", "employee", "destination", "purpose",
                     "trip_start", "trip_end", "days_claimed",
                     "total_cost", "predicted", "residual"]].round(2).to_string(index=False))
```

```
report_id   employee   destination        purpose trip_start   trip_end  days_claimed  total_cost  predicted  residual
  TR-2019  T. Nguyen  Columbus, OH Supplier audit 2026-01-27 2026-01-27             9      742.10    2943.17  -2201.07
  TR-2035    M. Chen Monterrey, MX Supplier audit 2026-02-02 2026-02-07             6     4185.60    2138.78   2046.82
  TR-2060    M. Chen Nashville, TN Customer visit 2026-04-21 2026-04-22             2     2968.75    1066.27   1902.48
  TR-2081 J. Alvarez    Denver, CO Sales training 2026-05-11 2026-05-15             5     3712.30    1870.65   1841.65
  TR-2084    R. Osei   Phoenix, AZ Customer visit 2026-06-03 2026-06-03             1     2500.00     798.14   1701.86
```

"You are an internal auditor at a building-hardware company. Using the table below of travel expense reports flagged by a regression model (cost vs. trip days), write a short memo to the Controller recommending which reports to approve, question, or reject, and why."

**CLAUDE**

## MEMORANDUM

**To:** Controller

**From:** Internal Audit

**Date:** October 4, 2026

**Re:** Travel expense reports flagged by the cost-vs-days regression model

### *Summary*

The model flagged five reports. I recommend approving one, questioning three, and rejecting one as submitted. The model predicts cost from trip length only, so a flag is a prompt for review, not evidence of wrongdoing. It cannot account for airfare, international travel, or conference fees.

| Report | Employee | Claimed | Predicted | Recommendation |
|---|---|---|---|---|
| TR-2019 | T. Nguyen | $742.10 | $2,943.17 | **Approve (correct the days field)** |
| TR-2035 | M. Chen | $4,185.60 | $2,138.78 | **Question** |
| TR-2060 | M. Chen | $2,968.75 | $1,066.27 | **Question (priority)** |
| TR-2081 | J. Alvarez | $3,712.30 | $1,870.65 | **Question** |
| TR-2084 | R. Osei | $2,500.00 | $798.14 | **Reject as submitted** |

### *Approve: TR-2019 (Nguyen, Columbus)*

This flag is a data error, not an overspend. The report shows a single-day trip (Jan 27) but claims 9 days. The model expected about $2,943 for nine days, so the large negative residual is an artifact. At $742 for one day, the cost is reasonable for a supplier audit. I recommend approving it and having the days field corrected so it doesn't distort future model runs.

### *Question*

- **TR-2060 (Chen, Nashville):** This claims $2,969 for two days, about $1,484 per day and nearly 3x the prediction, with no obvious reason for a domestic customer visit. It is also Chen's second flag, so I recommend requesting full itemization and receipts.
- **TR-2035 (Chen, Monterrey):** The trip is 6 days, matching the dates, and costs about 2x the prediction. International airfare could explain much of the gap, so this is likely approvable once receipts confirm it. I suggest reviewing it together with TR-2060, since the two reports total roughly $3,950 above predicted.
- **TR-2081 (Alvarez, Denver sales training):** The dates match the 5 days claimed, and the cost is about 2x the prediction. Registration or tuition fees may be included, and those should be coded separately from travel. I recommend requesting an itemized breakdown before approving.

### *Reject as submitted: TR-2084 (Osei, Phoenix)*

This claims exactly $2,500.00 for a one-day customer visit, over 3x the prediction and the highest cost per day in the set. A perfectly round total suggests an estimate or lump sum rather than actual receipts. I recommend returning it for resubmission with itemized receipts. This is a documentation rejection, not a finding of misconduct, and we should approve whatever the receipts support.

### *Exposure and next steps*

The four over-predicted reports total $13,366.65 claimed against $5,873.84 predicted, a gap of about $7,493. Not all of it will be unsupported, so the amount actually at risk depends on what the receipts show. I suggest a 10-business-day deadline for documentation. I also suggest a brief look at Chen's other recent claims, given two flags in one quarter, and a refresh of the model with destination and trip type (domestic vs. international) to reduce false positives.

**USER**

Can you give me a pdf of this chat with ur answer in it.